

BLG640E Security in Financial Information Systems

Spring 2026, Term Project

READ CAREFULLY ALL THE TEXT!

- i. Deadlines will **not be postponed**! Any email or request regarding postponing a deadline will not be answered!
- ii. Any material (your project report or any other...) sent **via email** will not be considered during grading!
- iii. Do **NOT** use **Generative AI** models to prepare your project. If we detect that some parts or all the project is prepared with Generative AI, you will get 0 point from the project. Note that your project reports will be analyzed with Turnitin. If Turnitin AI score > 20, you will get 0 from the project. This means that you may not collect enough points to pass from the course.
- iv. Do **NOT** put images, figures, or tables from other researches to your paper directly even if by referring them. You must create your own figures and tables.
- v. Do **NOT** read and cite papers/resources from open sources, such as arXiv! If your report contains reference(s) from open sources, you will get 0 points from the project.

Assignment 1: Determine the topic for your research paper

Due: 24.02.2026 (No extensions!)

You are expected to prepare a research paper about IoT Payments or IoT based Financial Systems such as :

- IoT Payments for E-Health Applications
- Security for IoT Financial Transactions
- Key Distribution for IoT Payments
- Privacy for IoT Border Payments
- Or suggest a new topic

Expected and recommended stages of the term project are as follows:

1. Determine your research topic.
2. Literature review and analysis.
3. Understand the problems in selected sub topic (application domain).
4. Select a research problem and explain the significance of the problem (you intend to solve)
5. Propose a solution related to the selected problem. The solution may be
 - a. Design and implementation for research paper.
 - b. Formal Analysis
 - c. ... simply new ideas! **DO NOT LIMIT YOUR IMAGINATION!**
6. Prove your solution for the research paper.
7. Present your term project
8. **YOUR SOLUTION MUST NOT CONTAIN ARTIFICIAL INTELLIGENCE (AI) OR MACHINE LEARNING (ML) BASED SOLUTION. SIMPLY DO NOT USE AI AND ML.**

What to submit? You should submit your selected research topic to **Ninova** only. You should submit your information in a pdf file.

Assignment 2: Prepare your report (30 pts) Due: 12.04.2026 (No extensions!)

You are expected to prepare the following parts of your paper in **IEEE conference paper format**, which is uploaded on Ninova under the name of **"conference-template-a4.docx"**. You must read sources including **journal papers** that have been published **since 2024**. You should carefully explain your understanding of papers you read **by using your own words/sentences (Use your own English even it is broken!)**.

Your research paper must have at least the following sections:

- 1) Title, authors, address of the author, email address (Author is you!)
- 2) Abstract (max 150 words)
 - I. Topic of your work
 - II. What are problems?
 - III. The significance of topic
 - IV. Contribution(s) of your work
- 3) Introduction (approximately 1 page)
 - I. What is the subject (Detailed explanation of the topic and problem)?
 - II. The importance of the subject?
 - III. What has been done in this paper?
 - IV. Contribution(s). (Your paper should have one to three contributions.)
- 4) Related Work (Find a specific name for the title of this section, such as "Key Distribution and IoT Payments ...") (between 1 and 1,5 pages)
 - I. Who has done work in this area?
 - II. Areas that inspired this work.
 - III. Motivate the current work (your work).
 - IV. Compare the existing works with table, charts, and/or other visual representations.
- 5) Proposed solution: (Write a specific name for your solution, which will closely be related to the topic of your paper, such as "Key Distribution for Privacy Preserving IoT Payments on E-Health") (between 1,5 and 2 pages)
 - I. You may explain the solution mathematically
 - II. Algorithmically,
 - III. Detailed innovative design, or
 - IV. In any other way.
 - V. **You should not cite any reference in this section.**
- 6) Analysis of your proposed solution (between 1,5 and 2,5 pages)
 - I. Empirically that means implementation.
 - II. Analytically (it may require advanced mathematics knowledge!)
 - III. Observational
 - IV. Or simply by comparisons.
- 7) Conclusion and Future Work (Problem, Solution, Contributions) (at most 0,5 page)
- 8) References: Your paper should contain references at least 15 (journal, conference, book, ...).

What to submit for Phase 1? You should submit your paper but not your presentation (in pdf format, not any other) to **Ninova** only. **Your paper should be at least 6 full pages long!**

Assignment 3: Prepare a presentation (after presentation 10 pts)

Prepare a presentation about your project. The presentations should contain the following information:

- Give general information about your topic
- Emphasize main problems
- Explain and defend the significance of your research paper
- Explain your solution and analysis
- Provide potential challenges
- Presentation should be **approximately 20 pages** (max 20 minutes for presentation + 5 minutes for questions). Depending on the number of students, the duration of presentations may be updated.

Note that you can update your presentation until the time slot you will receive!

What to submit? You will not submit or send presentations.

Do not send any material to me regarding the presentation!